

Financial Education

May 2026

Draft — Not for Circulation

Contents

Executive Summary	3
Research Agenda 1	4
1 Motivation and Research Question	4
2 Contribution to the Literature	6
3 Data and Setting	7
3.1 Partner Institutions	7
3.2 Student-Level Data	7
3.3 Instructor-Level Data	8
4 Experimental Design	8
4.1 The 2×2 Factorial Randomized Controlled Trial	8
4.2 Stratified Randomization	8
4.3 Implementation Protocol	9
5 Empirical Strategy	9
5.1 Step 1: Basic Intent-to-Treat Effects	9
5.2 Step 2: Quantile Treatment Effects	10
5.3 Step 3: Within-Class Variance Reduction	10
5.4 Step 4: Instructor Behavior Mechanism	11
5.5 Step 5: Heterogeneity by Class Size and Global South	12
5.6 Step 6: Long-Run Persistence	13
6 Power Analysis and Sample Size Determination	13
6.1 Effect Size Calibration	13
6.2 Required Sample Size	13
6.3 Pilot Protocol	14
7 Identification Challenges and Robustness	14
8 Expected Results and Scenarios	15
9 JEL Classification	16
10 Literature Review Strategy	16
10.1 The Five Literature Strands	17

10.1.1	Strand 1: The Teaching-to-the-Mean Problem and the 2-Sigma Hypothesis (Foundational)	17
10.1.2	Strand 2: Financial Literacy and Financial Education (Domain Motivation)	17
10.1.3	Strand 3: AI and EdTech as Equalizer (AI Application)	17
10.1.4	Strand 4: Technology Adoption and Inequality in Education (Distributional Context)	17
10.1.5	Strand 5: Instructor Information and Teacher Effectiveness (Mechanism Validation)	18
10.2	Critical Literature Gap: The Missing 2×2	18
11	Feasibility and Cost	18
12	Timeline	19
	How the Four Agendas Connect	20
	References	22

Executive Summary

The agenda applies analogous AI tools and distributional reasoning to a distinct but structurally parallel problem: the **teaching-to-the-mean failure** in financial education, where instructors can only serve the average student and must neglect both the strongest and the weakest learners.

Agenda 1: Flattening the Classroom Bell Curve. Deploys a dual-interface AI system in financial economics courses: one side guides students adaptively, the other sends differentiated early warnings to professors about students at the low tail (at risk), high tail (over-challenged), and the emerging middle. Tests whether the tool narrows within-class variance in financial literacy outcomes while improving the mean, with a factorial RCT design that identifies the separate and complementary effects of the student-side and teacher-side components. The Global South arm tests whether the tool is most effective precisely where class sizes are largest and preparation most heterogeneous.

The first three agendas form a coherent research program on peer review: Agenda 2 validates the AI tool, Agenda 1 deploys it to document the gender tax, and Agenda 3 tests whether the generative AI revolution has begun closing the gap. Agenda 4 extends the logic to a new domain: financial education in heterogeneous classrooms, where the structural problem—a gatekeeper who defaults to the average because they cannot individualize—is directly analogous.

Table 1: How Prior Proposals Map to the Four Agendas

Agenda	Incorporates
Agenda 1: Classroom Bell Curve	New agenda: dual-interface AI for financial education; 2×2 factorial RCT; distributional outcomes; Global South classroom heterogeneity

Research Agenda 1

Flattening the Classroom Bell Curve: An AI-Powered Dual-Interface System for Financial Education and the Early Warning Problem in Heterogeneous Classrooms

Target Journals: *Review of Financial Studies*, *Journal of Human Resources*, *American Economic Review*, *Applied Economics*

1 Motivation and Research Question

The teaching-to-the-mean problem. Lazear (2001) formalizes a fundamental constraint in education: an instructor must choose a single pace for the entire class. Teaching to the mean is individually rational for the instructor but collectively inefficient—students at the bottom of the distribution fall further behind, while students at the top are under-challenged and disengage. The instructor has neither the time nor the information to individualize attention at scale. This constraint is not a failure of effort or pedagogy: it is a structural feature of the classroom technology.

Bloom (1984) quantified the cost of this constraint in his landmark study of the “2-Sigma Problem”: students receiving one-on-one tutoring outperform those in conventional classrooms by approximately two standard deviations. Bloom called this difference a “problem” because one-on-one tutoring is unscalable. For forty years, the 2-Sigma Problem has defined the outer bound of what personalized instruction can achieve, and the inner bound of what conventional classrooms inevitably sacrifice.

Generative AI fundamentally changes the calculus on the student side—adaptive tools can deliver individualized content and feedback at near-zero marginal cost. But existing AI educational tools solve only half the problem. They ignore the information asymmetry on the *instructor’s* side: the professor still does not know, at sufficient resolution, which students need urgent attention, which are ready for enrichment, and which are invisibly disengaging from boredom rather than from difficulty.

The financial education context. Financial literacy is one of the most consequential and most heterogeneous domains of economic education. Lusardi and Mitchell (2014) document enormous variance in financial knowledge both within and across populations,

and link financial literacy deficits directly to inadequate retirement savings, excessive debt, and vulnerability to financial fraud. Introductory financial economics courses are the primary institutional vehicle for correcting these deficits—yet they face precisely the Lazear problem in its most acute form: students arrive with vastly different baseline knowledge (some have never maintained a bank account; others have managed investment portfolios), and a single lecture pace inevitably fails both tails.

The proposed solution. We deploy and evaluate a **dual-interface AI system** (the EduAlert System) in financial economics courses. The system has two complementary components:

1. **Student-facing interface (AdaptLearn):** Adaptive content delivery, personalized practice exercises, immediate formative feedback, and a dynamic difficulty adjustment engine. The student interacts with a conversational AI tutor that responds to their specific conceptual gaps, not to the class average.
2. **Instructor-facing interface (EarlyAlert Dashboard):** A real-time monitoring dashboard that classifies students into three alert categories and sends structured notifications to the instructor:
 - **Red alerts (at-risk):** Students exhibiting attendance gaps, persistent conceptual errors on foundational topics, declining engagement trends, or sudden drops in performance.
 - **Blue alerts (over-challenged/gifted):** Students with rapid completion times, high accuracy on all available material, and engagement signals consistent with under-stimulation. These students are at risk of disengagement from boredom, not difficulty.
 - **Yellow alerts (emerging concern):** Students who were stable but show early-warning signals toward either tail—a “watch list” that allows preemptive intervention before problems crystallize.

The key innovation is the **complementarity** between the two interfaces. The student tool improves learning conditional on a student engaging with it. The instructor dashboard solves the information problem that prevents the instructor from reallocating their scarce human attention toward the students who most need it. Neither component alone addresses both sides of the Lazear constraint.

Primary Research Question: Does a dual-interface AI system that provides personalized guidance to students and real-time differentiated early warnings to instructors reduce the within-class variance in financial literacy outcomes, while improving mean outcomes, relative to either component alone and to traditional instruction?

Secondary Research Questions:

1. Is the student-facing component (AdaptLearn) or the instructor-facing component (EarlyAlert) the primary driver of any variance reduction, and are they complements or substitutes?
2. Does the system produce differential benefits at the tails of the baseline ability distribution—larger gains for the bottom quartile from the alert system, and larger gains for the top quartile from the adaptive content?
3. Do instructors actually change their behavior in response to the EarlyAlert dashboard (office hours, personalized outreach, enrichment assignments), and does this behavioral change mediate the effect on learning outcomes?
4. Is the system more effective in larger, more heterogeneous classes—specifically, in Global South universities where class sizes are substantially larger and baseline variance in financial literacy is wider?
5. Do effects persist beyond the course? Does the system improve long-run financial literacy and downstream financial decisions?

2 Contribution to the Literature

First, this paper provides the first rigorous experimental test of whether AI tools can address the Lazear teaching-to-the-mean problem in higher education. Existing EdTech evaluations focus on **mean** learning outcomes. We focus on the **distribution** of outcomes—specifically on variance reduction and tail outcomes—which is the theoretically relevant quantity given the Lazear model. This repositions the evaluation question from “does AI help on average?” to “does AI flatten the bell curve?”

Second, the dual-interface design is methodologically novel. Prior AI education tools provide either a student-facing adaptive component or an instructor analytics dashboard, but not both in an integrated system evaluated by a randomized controlled trial. The 2×2 factorial design allows us to separately identify the *student-side effect* (adaptive learning), the *instructor-side effect* (attention reallocation), and their *interaction* (whether the two components are complements).

Third, this paper provides a direct test of the **2-Sigma hypothesis** (Bloom, 1984) in a real-world academic setting. Bloom’s finding was obtained in controlled tutoring experiments; whether AI-powered tools can approach the 2-sigma benchmark in naturalistic classroom settings is an open and important empirical question.

Fourth, by embedding the study in Global South universities, we test whether the tool is most effective precisely where the Lazear constraint is most binding—in classes

of 50–200 students with widely heterogeneous preparation. This provides both a test of external validity and a direct estimate of the distributional welfare impact of the technology in the context where it matters most.

Fifth, this agenda connects to the broader research program of Agendas 1–3 by applying the same logic of **AI as equalizer** to a different gatekeeping context. Just as peer reviewers default to a single standard (the gender tax), instructors default to a single pace (the teaching-to-the-mean tax). Both failures involve a gatekeeper who cannot individualize attention at scale, and both create predictable, systematic penalties for those at the tails of the distribution.

3 Data and Setting

3.1 Partner Institutions

We partner with universities offering introductory financial economics courses, stratified by:

- **Geographic tier:** Core institutions (Western Europe, North America) and Global South institutions (Latin America, Sub-Saharan Africa, South/Southeast Asia).
- **Class size:** Small ($N < 35$ students), medium ($N = 35\text{--}80$), large ($N > 80$). Global South courses typically fall in the medium-to-large range.
- **Instructor experience:** Junior (first 3 years of teaching) vs. senior faculty.

3.2 Student-Level Data

For each student i in class j :

- **Financial literacy assessments:** Pre-course (baseline), mid-semester, and end-of-course scores using a validated instrument adapted from [Lusardi and Mitchell \(2014\)](#), covering compound interest, diversification, inflation, risk, and financial planning.
- **Platform engagement logs** (treatment arms only): Login frequency, time-on-task, exercise completion rates, error patterns, speed-accuracy profiles, and adaptive difficulty level reached.
- **Administrative records:** Attendance, prior GPA, declared major, year of study, and demographic characteristics.
- **Long-run follow-up:** Financial literacy re-assessment at 6 months post-course; self-reported financial behavior (savings rate, investment participation, debt management).

3.3 Instructor-Level Data

For each instructor j :

- **Dashboard interaction logs** (T2 and T3 arms): Alert acknowledgment rates, alert-to-response time (did the instructor follow up within k days?), type of response taken (office hours, email, additional resources).
- **Instructor attention allocation**: Office hours attendance sheets, number of personalized emails sent, time spent per student quartile (self-reported via teaching diary).
- **Teaching experience and beliefs**: Pre-study survey on beliefs about within-class heterogeneity and their capacity to address it.

4 Experimental Design

4.1 The 2×2 Factorial Randomized Controlled Trial

We randomize at the **classroom level** (to prevent within-class spillovers and social desirability effects) using a 2×2 factorial design:

Table 2: 2×2 Factorial Treatment Design

	No EarlyAlert Dashboard	EarlyAlert Dashboard
No AdaptLearn	Control (C)	T_2 (Dashboard only)
AdaptLearn	T_1 (Student AI only)	T_3 (Full system)

This design allows us to identify:

- **Student-side effect**: $T_1 - C$ (adaptive learning, no instructor information)
- **Instructor-side effect**: $T_2 - C$ (instructor dashboard, no student AI tool)
- **Interaction/complementarity**: $T_3 - T_1 - T_2 + C$ (does the full system produce more than the sum of its parts?)

4.2 Stratified Randomization

Classroom-level stratification on: (a) institution geographic tier (Core vs. Global South); (b) class size (small, medium, large); (c) instructor seniority; (d) mean baseline financial literacy score of enrolled students; (e) baseline within-class variance in financial literacy.

4.3 Implementation Protocol

1. **Weeks 1–2:** Baseline financial literacy assessment administered to all students across all arms. EduAlert system activated for T1, T2, T3 arms.
2. **Weeks 3–14 (mid-semester):** AdaptLearn active for T1 and T3 students. EarlyAlert Dashboard active for T2 and T3 instructors, with weekly alert summaries.
3. **Week 8:** Mid-semester financial literacy re-assessment (all arms). Used for alert calibration in T2/T3 and for interim power analysis.
4. **Week 16:** End-of-course financial literacy assessment (primary outcome).
5. **6-month follow-up:** Financial literacy and financial behavior survey.

5 Empirical Strategy

5.1 Step 1: Basic Intent-to-Treat Effects

We estimate intent-to-treat (ITT) effects using ordinary least squares with pre-specified controls:

$$y_{ij} = \alpha + \beta_1 T_{1j} + \beta_2 T_{2j} + \beta_3 T_{3j} + \gamma y_{ij,0} + \mathbf{\Gamma} \mathbf{W}_{ij} + \mathbf{\Pi} \mathbf{Z}_j + \varepsilon_{ij} \quad (1)$$

where:

y_{ij} is the end-of-course financial literacy score of student i in classroom j , standardized to mean zero and unit variance within the control group.

T_{1j}, T_{2j}, T_{3j} are indicators for whether classroom j was assigned to the student-AI-only, dashboard-only, or full-system arm.

$y_{ij,0}$ is the baseline financial literacy score (pre-registered covariate), included for precision and to control for potential baseline imbalance.

$\mathbf{W}_{ij}$ is a vector of student-level controls: year of study, prior GPA, declared major, gender, and attendance rate.

$\mathbf{Z}_j$ is a vector of classroom-level controls: class size, instructor experience, institution type (Core/Global South), and stratification cell fixed effects.

ε_{ij} is the error term. Standard errors are clustered at the classroom level j throughout.

$\beta_1, \beta_2, \beta_3$ are the ITT effects of each treatment arm on the mean financial literacy outcome.

The complementarity parameter: We compute $\hat{\beta}_3 - \hat{\beta}_1 - \hat{\beta}_2$ and test whether it is significantly positive (superadditive complementarity between the two interfaces) or negative (substitutability). A positive value means the full system produces more than the sum of its parts—the student tool and the instructor tool reinforce each other.

5.2 Step 2: Quantile Treatment Effects

The primary theoretical prediction concerns the tails of the distribution, not the mean. We estimate quantile treatment effects for each treatment arm:

$$Q_\tau(y_{ij} \mid T_{1j}, T_{2j}, T_{3j}, \mathbf{W}_{ij}) = \alpha_\tau + \beta_{1\tau}T_{1j} + \beta_{2\tau}T_{2j} + \beta_{3\tau}T_{3j} + \gamma_\tau y_{ij,0} + \mathbf{\Gamma}_\tau \mathbf{W}_{ij} + \varepsilon_{ij}(\tau) \quad (2)$$

where:

$Q_\tau(\cdot)$ is the τ -th conditional quantile of the outcome distribution, estimated for $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$.

$\beta_{k\tau}$ is the effect of treatment k at quantile τ of the outcome distribution.

Key hypotheses:

- (H2a) **Bottom-tail lifting:** $\beta_{2,\tau=0.10} > 0$ and $\beta_{3,\tau=0.10} > 0$. The EarlyAlert Dashboard helps the bottom tail by directing instructor attention toward at-risk students.
- (H2b) **Top-tail advancement:** $\beta_{1,\tau=0.90} > 0$ and $\beta_{3,\tau=0.90} > 0$. The AdaptLearn component helps the top tail by providing enrichment content that traditional instruction cannot deliver.
- (H2c) **Variance compression:** The treatment effects are larger at the tails than at the median— $|\beta_{k,0.10}| > |\beta_{k,0.50}|$ and $|\beta_{k,0.90}| > |\beta_{k,0.50}|$. This is the central test of the Lazear flattening hypothesis.

5.3 Step 3: Within-Class Variance Reduction

We construct classroom-level measures of outcome variance and estimate whether treatment reduces them:

$$\hat{\sigma}_j^2 = \alpha + \theta_1 T_{1j} + \theta_2 T_{2j} + \theta_3 T_{3j} + \mathbf{\Pi} \mathbf{Z}_j + \nu_j \quad (3)$$

where:

$\hat{\sigma}_j^2$ is the estimated within-class variance in the end-of-course financial literacy score for classroom j , residualized on baseline scores to remove the mechanical correlation between variance and means.

θ_k is the effect of treatment arm k on within-class score dispersion.

$\mathbf{Z}_j$ includes the baseline within-class variance (to control for mean reversion in variance) and all stratification variables.

Hypothesis 3 (Variance Compression): If $\theta_3 < 0$ and $|\theta_3| > |\theta_1| + |\theta_2|$, the full system compresses within-class dispersion beyond the sum of its components—direct evidence of the complementarity between the student and instructor interfaces in addressing the Lazear constraint.

We also test whether variance reduction is driven primarily by bottom-tail improvement (pulling up the weakest students) or top-tail improvement (advancing the strongest students) by decomposing $\hat{\sigma}_j^2$ into contributions from different parts of the distribution.

5.4 Step 4: Instructor Behavior Mechanism

We test whether the EarlyAlert Dashboard changes instructor behavior, and whether this behavioral change mediates the effect on student outcomes (for T2 and T3 classrooms):

$$M_{ij} = \alpha + \rho_1 T_{2j} + \rho_2 T_{3j} + \gamma \mathbf{AlgoAlert}_{ij} + \mathbf{\Gamma W}_{ij} + \delta_j + \varepsilon_{ij} \quad (4)$$

$$y_{ij} = \alpha + \beta_1 T_{1j} + \beta_2 T_{2j} + \beta_3 T_{3j} + \kappa M_{ij} + \gamma y_{ij,0} + \mathbf{\Gamma W}_{ij} + \varepsilon_{ij} \quad (5)$$

where:

M_{ij} is a measure of instructor-directed attention toward student i : the number of personalized outreach events (emails, office hours visits, additional material shared) recorded during the semester.

$\mathbf{AlgoAlert}_{ij}$ is the algorithmic alert category assigned to student i by the EarlyAlert system (Red/Blue/Yellow/None). Included to test whether instructors respond differentially to different alert types.

ρ_1, ρ_2 capture whether the dashboard increases instructor attention directed to specific students (the first-stage mechanism).

κ captures whether instructor attention improves outcomes, conditional on treatment (the second stage). We instrument M_{ij} with T_{2j} and T_{3j} to address endogeneity of instructor attention.

δ_j are classroom fixed effects (in the first-stage equation only), absorbing all class-level determinants of instructor behavior.

Hypothesis 4 (Attention Reallocation): If $\rho_1 > 0$ (or $\rho_2 > 0$), the dashboard increases instructor attention toward students who receive alerts, compared to non-alerted students. This is the mechanism through which the instructor interface should operate.

We further test whether instructor responses differ by alert type: do instructors respond more urgently to Red (at-risk) alerts than Blue (gifted) alerts? An asymmetric response would indicate an attentional bias toward the bottom tail, which may be welfare-improving if the returns to remedial attention exceed the returns to gifted education—or welfare-reducing if over-challenged students are systematically neglected.

5.5 Step 5: Heterogeneity by Class Size and Global South

We test whether the full system (T_3) is more effective in larger, more heterogeneous classrooms—the key theoretical prediction from the Lazear model, since the welfare loss from teaching to the mean is larger when the within-class variance is larger:

$$y_{ij} = \alpha + \beta_1 T_{3j} + \beta_2 \text{LargeClass}_j + \beta_3 \text{GS}_j + \beta_4 (T_{3j} \times \text{LargeClass}_j) + \beta_5 (T_{3j} \times \text{GS}_j) + \beta_6 (T_{3j} \times \text{HighVariance}_{j,0}) + \gamma y_{ij,0} + \mathbf{\Gamma W}_{ij} + \varepsilon_{ij} \quad (6)$$

where:

LargeClass_j is an indicator for classrooms with $N > 80$ students, where the teaching-to-the-mean constraint is most binding.

GS_j is an indicator for Global South institutions.

$\text{HighVariance}_{j,0}$ is an indicator for classrooms in the top tercile of baseline within-class variance in financial literacy scores.

β_4 tests whether the treatment effect is amplified in large classes: the Lazear amplification hypothesis.

β_5 tests whether Global South classrooms see larger gains, consistent with the larger variance and weaker institutional support in those settings.

β_6 tests whether the tool is most effective in the classrooms with the highest ex-ante heterogeneity.

This is the paper’s key Global South contribution: if $\beta_4 > 0$ and $\beta_5 > 0$, the technology delivers the largest welfare gains precisely where the conventional classroom most fails its students—providing a direct economic rationale for prioritizing deployment in large, heterogeneous, under-resourced settings.

5.6 Step 6: Long-Run Persistence

We assess whether improvements in financial literacy and financial behavior persist six months after the end of the course:

$$y_{ij,+6} = \alpha + \sum_{k=1}^3 \beta_k^{+6} T_{kj} + \gamma^{+6} y_{ij,0} + \phi y_{ij,\text{end}} + \mathbf{\Gamma W}_{ij} + \varepsilon_{ij} \quad (7)$$

where:

$y_{ij,+6}$ is the financial literacy score six months post-course (or a financial behavior index, combining savings, investment, and debt management outcomes).

$y_{ij,\text{end}}$ is the end-of-course score, included to distinguish between: (a) persistent effects of the educational intervention on underlying knowledge, and (b) test-taking improvements that fade. The parameter of interest is β_k^{+6} conditional on $y_{ij,\text{end}}$ —whether students in treated classrooms retain more knowledge over time, not just perform better immediately.

Hypothesis 6 (Persistence): If $\beta_3^{+6} > \beta_1^{+6}$ (the full system has more persistent effects than the student-AI-alone), the instructor attention reallocation channels produce deeper and more durable learning, not merely better performance on the end-of-course assessment.

6 Power Analysis and Sample Size Determination

6.1 Effect Size Calibration

Noy and Zhang (2023) find that AI writing assistance improves quality by 0.3–0.5 SD in their RCT, with the largest gains for the weakest writers. VanLehn (2011) meta-analyzes intelligent tutoring systems and reports effect sizes of 0.3–1.0 SD relative to conventional instruction, depending on the implementation. For our context, we adopt a conservative main effect of $\delta = 0.25$ SD on mean financial literacy outcomes, and a variance compression of 20–30% for the full system.

6.2 Required Sample Size

For the mean outcome (Step 1), with classroom-level randomization, an intra-class correlation of $\rho_{\text{ICC}} = 0.10$ (typical for educational outcomes), and 30 students per classroom:

$$N_{\text{classrooms}} = \frac{2(z_{\alpha/2} + z_{\beta})^2 [1 + (n-1)\rho_{\text{ICC}}]}{n \delta^2} \quad (8)$$

where $n = 30$ is the average class size and $\delta = 0.25$ SD:

Table 3: Required Classrooms by Arm ($\alpha = 0.05$, Power = 0.80, $n = 30$, ICC = 0.10)

Parameter	δ (SD)	Classrooms/arm	Total classrooms
Mean outcome (β_k , OLS)	0.25	22	88
Variance compression (θ_k)	0.25 reduction	28	112
Quantile effect at $\tau = 0.10$ ($\beta_{k,0.10}$)	0.30	18	72
Heterogeneity: GS $\times$ T3 (β_5)	0.35	24	96

Target sample: 120 classrooms (30 per arm), with approximately 30 students per classroom, yielding $N \approx 3,600$ students total. Oversample Global South classrooms (target: 40% Global South, 60% Core) to ensure the heterogeneity analysis in Step 5 is well-powered.

6.3 Pilot Protocol

Before full deployment: (1) implement the system in 8 classrooms (2 per arm) for one semester; (2) estimate the ICC and the preliminary effect size; (3) assess instructor compliance with the EarlyAlert dashboard; (4) refine alert thresholds based on pilot alert rates.

7 Identification Challenges and Robustness

Classroom-level spillovers. Students from different arms may interact. Mitigation: randomize at the classroom level; avoid multiple treatment arms within the same course section.

Instructor compliance with the dashboard. If instructors in T2/T3 do not engage with the EarlyAlert alerts, the dashboard has no effect by construction. We pre-register the compliance threshold (instructor must acknowledge $\geq 50\%$ of alerts); conduct compliance analysis; and estimate both ITT and complier-average causal effects (CACE) using alert acknowledgment as the first stage.

Hawthorne effects. Instructors in the dashboard arm may change behavior merely from awareness that they are being observed. Mitigation: all instructors are informed that the study measures student outcomes, not instructor behavior; the dashboard is framed as a teaching support tool, not a monitoring device.

Differential attrition. Students in the treatment arms may be more or less likely to complete the end-of-course assessment. We test for differential attrition by treatment status and apply inverse probability weighting if attrition is non-random.

Multiple testing. With 3 treatment comparisons $\times$ 5 quantiles $\times$ 6 steps, we pre-register primary and secondary hypotheses; apply Benjamini-Hochberg correction for secondary analyses; and treat the variance compression test (Step 3) as the pre-registered primary outcome.

Robustness checks:

1. Alternative primary outcomes: mid-semester score, financial behavior index at 6-month follow-up.
2. Alternative ICC assumptions and cluster-robust standard errors.
3. Restricting to classrooms where instructor compliance exceeds the pre-registered threshold.
4. Excluding classrooms where baseline balance tests fail at $p < 0.10$.
5. Within-instructor regressions for instructors who teach multiple sections across arms.

8 Expected Results and Scenarios

Scenario A: Full Complementarity. T_3 produces variance compression significantly larger than $T_1 + T_2 - C$; bottom-tail and top-tail effects are both significant and larger than median effects; instructor attention reallocation mediates the T_2 and T_3 effects; Global South classrooms show larger treatment effects consistent with higher baseline variance. *Implication:* deploy the dual-interface system as a standard teaching technology, with priority for large, heterogeneous Global South classrooms.

Scenario B: Student-Side Dominance. $T_1 \approx T_3 \gg T_2 \approx C$; variance compression driven by AdaptLearn, not by the dashboard; instructor behavior does not change significantly in T2/T3. *Implication:* the bottleneck is student access to adaptive content, not instructor information. AI tutors alone can address the Lazear problem without requiring instructor behavior change—a scalable finding that lowers the barrier to deployment.

Scenario C: Instructor Dashboard Dominance for the Bottom Tail. T_2 produces significant bottom-tail gains but minimal top-tail or mean gains; T_1 produces significant top-tail gains but no bottom-tail effect; T_3 produces both. *Implication:* the two components address different tails of the distribution and are complements in the sense that a system targeting only one tail leaves the other unaddressed. This has direct implications for prioritization: institutions concerned primarily with at-risk students should emphasize the dashboard; institutions concerned with gifted students should emphasize AdaptLearn.

Scenario D: Null Results. All treatment effects near zero on variance; mean effects modest and driven by demand-side factors (student effort). *Implication:* the Lazear constraint is more resistant to technological solutions than the 2-Sigma hypothesis predicts. Deeper institutional reform—tracking, differentiated curricula, smaller classes—may be necessary. This is itself a high-value finding that clarifies the limits of AI-based EdTech.

9 JEL Classification

Table 4: JEL Code Classification for Agenda 4

JEL Code	Category	Relevance
G53	Financial Literacy	Core: financial education as primary outcome domain
I21	Analysis of Education	Teaching-to-the-mean problem; 2-Sigma hypothesis
I24	Education and Inequality	Distributional outcomes; tail effects; heterogeneity
C93	Field Experiments	2×2 factorial RCT design
O33	Technological Change	AI as equalizing technology
O12	Microeconomic Development	Global South class size and heterogeneity
J24	Human Capital; Skills	Financial literacy as human capital
C55	Large Data Sets: Modeling and Analysis	Platform engagement data; adaptive AI engine

Primary fields (Applied Economics): *Labor & Demographic Economics* (I21, I24, J24) and *Mathematical & Quantitative Methods* (C93, C55). **Secondary:** *Financial Economics* (G53) and *Development & Innovation* (O33, O12).

10 Literature Review Strategy

10.1 The Five Literature Strands

10.1.1 Strand 1: The Teaching-to-the-Mean Problem and the 2-Sigma Hypothesis (Foundational)

Key papers: Lazear (2001) (formalization of the optimal classroom problem; the trade-off between class size, teaching pace, and within-class variance); Bloom (1984) (the 2-Sigma Problem; one-on-one tutoring vs. conventional instruction); Duflo et al. (2011) (tracking and peer effects in Kenya; Duflo, Dupas, and Kremer show that ability grouping can reduce the within-classroom heterogeneity that drives Lazear’s mechanism). *Positioning:* we test whether AI can solve the 2-Sigma Problem at scale, complementing but not requiring the tracking solution that Duflo et al. evaluate.

10.1.2 Strand 2: Financial Literacy and Financial Education (Domain Motivation)

Key papers: Lusardi and Mitchell (2014) (the economic importance of financial literacy; the canonical measurement framework); Lusardi and Mitchell (2011) (financial literacy around the world; documents the within-country variance that motivates our intervention); Banerjee et al. (2007) (Pratham remedial education RCT; the closest antecedent in education RCTs for at-risk learners). *Positioning:* we embed the financial literacy intervention in a rigorous distributional framework, shifting the evaluation question from “does financial education work?” (Lusardi and Mitchell, 2014) to “does it flatten the distribution?”

10.1.3 Strand 3: AI and EdTech as Equalizer (AI Application)

Key papers: VanLehn (2011) (meta-analysis of intelligent tutoring systems; establishes effect-size benchmarks for AI tutors); Muralidharan and Sundararaman (2015) (personalized technology-aided instruction in India; direct antecedent for adaptive learning in developing countries); Brynjolfsson et al. (2023) and Noy and Zhang (2023) (AI compresses the performance distribution; carried over from Agendas 1–3). *Positioning:* we extend the AI-as-equalizer literature from professional settings to higher education, and test for the first time whether the distributional compression documented by Brynjolfsson et al. (2023) extends to the within-classroom distribution of financial literacy.

10.1.4 Strand 4: Technology Adoption and Inequality in Education (Distributional Context)

Key papers: Acemoglu and Restrepo (2020) (technology adoption and the distribution of gains; the framework for thinking about who benefits from AI tools); Muralidharan and Sundararaman (2015, technology and educational inequality in India); Hjort and Poulsen

(2019, fast internet and human capital in Africa). *Positioning:* we test the specific prediction that AI tools are most equalizing when baseline heterogeneity is largest—implying that Global South classrooms should benefit disproportionately if the technology reaches them.

10.1.5 Strand 5: Instructor Information and Teacher Effectiveness (Mechanism Validation)

Key papers: Chetty, Friedman, and Rockoff (2014, teacher value-added and long-run student outcomes; establishes that instructor quality matters enormously and is heterogeneous); Kane and Staiger (2008, measuring teacher effectiveness); Duflo et al. (2011) again (teacher incentives and tracking). *Positioning:* we test whether giving instructors better information—the EarlyAlert dashboard—allows them to act more like the high-value-added teachers in Chetty et al., not by changing their intrinsic ability but by resolving the information asymmetry that prevents them from allocating attention optimally.

10.2 Critical Literature Gap: The Missing 2×2

Table 5: Literature Gap: Student AI $\times$ Instructor Information

	Instructor has information	Instructor does not have information
Student has AI tool	This paper (T_3)	Existing EdTech (T_1)
Student has no AI tool	This paper (T_2)	Traditional classroom (C)

Existing EdTech evaluations sit exclusively in the bottom-right cell (traditional classroom) or the top-right cell (student AI, no instructor information). The top-left cell—instructor with information, student without AI tool—is an underexplored but policy-relevant arm: it approximates what happens when instructors are given better administrative data systems without any AI student tool. The bottom-left cell—instructor information alone—is almost entirely unstudied in a randomized setting.

11 Feasibility and Cost

A major advantage of this design is that it builds on existing institutional relationships developed for Agendas 1–3 (partner universities, AI infrastructure, author classification pipelines) and extends them to an experimental setting. The EduAlert System is a new deployment of the ResearchAI multi-agent infrastructure, adapted from evaluating research papers to evaluating student learning.

Table 6: Estimated Budget for Agenda 4

Item	Cost	Notes
EduAlert System development and adaptation	\$15,000–25,000	One-time; leverages existing ResearchAI
Platform licensing and server costs	\$8,000–12,000	Per semester of deployment
Partner institution coordination	\$10,000–15,000	Travel, RA time, MoU negotiation
Financial literacy assessment instruments	\$2,000–4,000	Licensing and translation
Research assistant (data pipeline)	\$20,000–30,000	2 RAs for 12 months
6-month follow-up survey	\$5,000–8,000	Online survey, reminder costs
Total estimated budget	\$60,000–94,000	

Unlike Agendas 1–3, Agenda 4 requires IRB approval, partner institution agreements, and coordination with instructors. The timeline is therefore longer and depends on institutional partnership development. However, the marginal cost of adding the Global South arm is low if partnerships are established at the pilot stage.

12 Timeline

Table 7: Timeline for Agenda 4

Phase	Activity	Timeline
1	IRB approval and partner institution MoUs	Months 1–3
2	EduAlert System development and adaptation from ResearchAI	Months 2–5
3	Financial literacy instrument translation and validation	Months 3–5
4	Pilot study (8 classrooms, 2 per arm, 1 semester)	Months 5–10
5	Pilot analysis: ICC, effect size, compliance rates	Month 11
6	Full deployment: 120 classrooms, 4 arms	Months 12–22
7	End-of-course assessment and primary data collection	Month 22
8	6-month follow-up survey	Month 28
9	Main analysis (Steps 1–6)	Months 22–28
10	Robustness checks and heterogeneity analysis	Months 27–30
11	Paper writing and submission	Months 29–34

How the Four Agendas Connect

The four agendas form a two-tier research program. The first tier (Agendas 1–3) addresses the peer review problem. The second tier (Agenda 4) addresses an analogous but structurally distinct gatekeeping failure in education.

Tier 1: The Peer Review Program (Agendas 1–3).

1. **Agenda 2 validates the tool.** Before using AI as a gender-blind benchmark (Agenda 1) or interpreting AI-driven quality changes (Agenda 3), we must verify that the AI itself does not discriminate. Agenda 2 answers this question with a rigorous audit study.
2. **Agenda 1 documents the problem.** Using the validated AI tool, Agenda 1 quantifies the gender tax across 12 dimensions, identifies where women bear the heaviest burden, and tests whether the tax compounds for Global South researchers.
3. **Agenda 3 tests whether the problem is changing.** Exploiting the ChatGPT/GPT-4 natural experiment, Agenda 3 tests whether the gender tax has begun to narrow, adding time-series variation to Agenda 1's cross-sectional baseline.

Tier 2: The Classroom Problem (Agenda 4). Agenda 4 extends the core logic to a new domain: financial education in heterogeneous classrooms. The structural parallel is exact—an instructor who defaults to the average student creates the same kind of systematic penalty for those at the tails as a peer reviewer who defaults to a single standard creates for women and Global South researchers. In both cases, the gatekeeper cannot individualize attention at scale, and AI offers a potential solution.

Four conceptual bridges connect the two tiers:

- **The equalization hypothesis.** Agendas 3 and 4 both test whether AI tools close gaps that human institutions create. Agenda 3 tests this in the time dimension (before vs. after the AI shock); Agenda 4 tests this in the cross-sectional dimension (treated vs. control classrooms).
- **The Global South dimension.** All four agendas test whether the benefits of AI tools reach Global South researchers and students. The expected gradient—larger benefits where institutional support is weakest—is the same in all four agendas.

- **The distributional focus.** Agendas 1 and 3 focus on the distribution of peer review burdens (women at one tail, men at the other). Agenda 4 focuses on the distribution of learning outcomes within classrooms. Both reframe the evaluation question from mean effects to distributional effects.
- **Shared AI infrastructure.** The ResearchAI multi-agent system is the backbone of Agendas 1–3. Agenda 4 adapts the same infrastructure—multi-agent evaluation, real-time scoring, structured feedback—for the classroom context. Building the infrastructure for one agenda makes the others substantially cheaper at the margin.

Realistic consolidated timeline:

- **Months 1–2:** Pilots for Agendas 1, 2, and 4 simultaneously. IRB initiated for Agenda 4.
- **Months 2–8:** Data collection for Agendas 1–3 (shared pipeline). Agenda 2 audit study. IRB and partner MoUs finalized for Agenda 4.
- **Months 5–10:** Agenda 4 pilot semester (8 classrooms). Agenda 2 paper drafted.
- **Months 8–12:** Agenda 1 and 3 analysis. Agenda 4 pilot analysis and system refinement.
- **Months 12–22:** Agenda 4 full deployment (120 classrooms). Agendas 1 and 3 drafted and submitted.
- **Months 22–34:** Agenda 4 data collection completed (including 6-month follow-up), analysis, and submission.

Total realistic timeline: 34 months from pilot to submission of all four papers. Agendas 1–3 submit within 17 months. Agenda 4 requires the additional time for IRB, deployment, and follow-up. The four papers share infrastructure, theory, and the Global South thread, ensuring that the program as a whole is greater than the sum of its parts.

References

- Acemoglu, D. and Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy*, 128(6):2188–2244.
- Alexander, D., Gorelkina, O., Hengel, E., and Tol, R. (2023). Gender and the time cost of peer review. *Working paper*.
- Banerjee, A., Cole, S., Duflo, E., and Linden, L. (2007). Remedying education: Evidence from two randomized experiments in India. *Quarterly Journal of Economics*, 122(3):1235–1264.
- Bayer, P. and Charles, K. (2018). Divergent paths: A new perspective on earnings differences between black and white men since 1940. *Quarterly Journal of Economics*, 133(3):1459–1501.
- Bertrand, M. and Mullainathan, S. (2004). Are Emily and Greg more employable than Lakisha and Jamal? A field experiment on labor market discrimination. *American Economic Review*, 94(4):991–1013.
- Biber, D. and Gray, B. (2016). *Grammatical Complexity in Academic English: Linguistic Change in Writing*. Cambridge University Press.
- Blank, R. (1991). The effects of double-blind versus single-blind reviewing: Experimental evidence from the *American Economic Review*. *American Economic Review*, 81(5):1041–1067.
- Bloom, B.S. (1984). The 2 sigma problem: The search for methods of group instruction as effective as one-to-one tutoring. *Educational Researcher*, 13(6):4–16.
- Bohren, J.A., Imas, A., and Rosenberg, M. (2019). The dynamics of discrimination: Theory and evidence. *American Economic Review*, 109(10):3395–3436.
- Brock, J.M. and De Haas, R. (2022). Discriminatory lending: Evidence from bankers in the lab. *American Economic Journal: Applied Economics*, 15(2):31–68.
- Brynjolfsson, E., Li, D., and Raymond, L. (2023). Generative AI at work. *NBER Working Paper* 31161.
- Card, D., DellaVigna, S., Funk, P., and Iriberry, N. (2020). Are referees and editors in economics gender neutral? *Quarterly Journal of Economics*, 135(1):269–327.
- Checco, A., Bracciale, L., Loreti, P., Pinfield, S., and Bianchi, G. (2021). AI-assisted peer review. *Humanities and Social Sciences Communications*, 8(1):1–11.

- Chetty, R., Friedman, J., and Rockoff, J. (2014). Measuring the impacts of teachers II: Teacher value-added and student outcomes in adulthood. *American Economic Review*, 104(9):2633–2679.
- Coffman, K. (2014). Evidence on self-stereotyping and the contribution of ideas. *Quarterly Journal of Economics*, 129(4):1625–1660.
- Das, J., Do, Q.-T., Shaines, K., and Srinivasan, S. (2013). U.S. and them: The geography of academic research. *Journal of Development Economics*, 105:112–130.
- Dressel, J. and Farid, H. (2018). The accuracy, fairness, and limits of predicting recidivism. *Science Advances*, 4(1):eaao5580.
- Duflo, E., Dupas, P., and Kremer, M. (2011). Peer effects, teacher incentives, and the impact of tracking: Evidence from a randomized evaluation in Kenya. *American Economic Review*, 101(5):1739–1774.
- Ellison, G. (2002). The slowdown of the economics publishing process. *Journal of Political Economy*, 110(5):947–993.
- Exley, C. and Kessler, J. (2022). The gender gap in self-promotion. *Quarterly Journal of Economics*, 137(3):1345–1381.
- Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3):535–574.
- Hengel, E. (2022). Publishing while female: Are women held to higher standards? Evidence from peer review. *Economic Journal*, 132(648):2951–2991.
- Hjort, J. and Poulsen, J. (2019). The arrival of fast internet and employment in Africa. *American Economic Review*, 109(3):1032–1079.
- Hyland, K. (2005). Stance and engagement: A model of interaction in academic discourse. *Discourse Studies*, 7(2):173–192.
- Kleinberg, J., Lakkaraju, H., Leskovec, J., Ludwig, J., and Mullainathan, S. (2018). Human decisions and machine predictions. *Quarterly Journal of Economics*, 133(1):237–293.
- Kline, P., Rose, E., and Walters, C. (2022). Systemic discrimination among large U.S. employers. *Quarterly Journal of Economics*, 137(4):1963–2036.
- Lazear, E.P. (2001). Educational production. *Quarterly Journal of Economics*, 116(3):777–803.

- Liang, W., Yuksekgonul, M., Mao, Y., Wu, E., and Zou, J. (2023). GPT detectors are biased against non-native English writers. *Patterns*, 4(7):100779.
- Lusardi, A. and Mitchell, O.S. (2011). Financial literacy around the world: An overview. *Journal of Pension Economics and Finance*, 10(4):497–508.
- Lusardi, A. and Mitchell, O.S. (2014). The economic importance of financial literacy: Theory and evidence. *Journal of Economic Literature*, 52(1):5–44.
- Muralidharan, K. and Sundararaman, V. (2015). The aggregate effect of school choice: Evidence from a two-stage experiment in India. *Quarterly Journal of Economics*, 130(3):1011–1066.
- Noy, S. and Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654):187–192.
- Obermeyer, Z., Powers, B., Vogeli, C., and Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464):447–453.
- Raghavan, M., Barocas, S., Kleinberg, J., and Levy, K. (2020). Mitigating bias in algorithmic hiring: Evaluating claims and practices. *ACM Conference on Fairness, Accountability, and Transparency*, 469–481.
- Sarsons, H., Gerarden, K., Cranmer, P., and Heyes, A. (2021). Gender differences in recognition for group work. *Journal of Political Economy*, 129(1):101–147.
- VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist*, 46(4):197–221.